

The tree of rationals generated by $x \mapsto x + 1$ and $x \mapsto -1/x$: an exact distance formula and the rank recurrence

A solution to Problem 137 of Kagey’s Open Problem Collection

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August 2026

Abstract

Start at 0 and keep applying $f(x) = x + 1$ and $g(x) = -1/x$. Every rational number shows up eventually, and sorting them by the first step at which they appear gives the tree in OEIS A226247/A226248. Let $a(n)$ be the number of rationals that first appear after n steps. Kagey asks whether $a(n) = a(n - 1) + a(n - 3)$ for $n \geq 4$, whether the vertices reached last by g are exactly the negative ones, and whether the rationals at a given step can be described directly.

We answer all 3 questions. The main tool is a formula for the rank $d(x)$ of a rational x , the step at which it first appears. We define a *parent map* by $P(x) = x - 1$ for $x > 0$ and $P(x) = -1/x$ for $x < 0$, and show that it takes x back to 0 in exactly $d(x)$ steps. If $x > 0$ has negative (Hirzebruch–Jung) continued fraction $x = [c_1, \dots, c_k]^-$, this gives $d(x) = c_1 + \dots + c_k + k - 1$, and there is a similar formula with the ordinary continued fraction. From here the tree structure follows. The generating function of $a(n)$ is $(1 + t^2 - t^3)/(1 - t - t^3)$, so the recurrence holds for $n \geq 4$ and fails at $n = 3$. A vertex is reached last by g exactly when it is negative, and the rationals of rank n correspond to compositions of integers. This also fills the gap in the accepted Math.SE answer, which counted words in f and g but did not show that words and rationals match up one to one.

1 Introduction

In 2013 Kimberling [4] observed that the rational numbers can be generated from 0 by the two maps

$$f(x) = x + 1, \quad g(x) = -\frac{1}{x},$$

and organized them into generations. Generation 1 is (0), and generation $n + 1$ is made of the values $c + 1$ and $-1/c$ (for $c \neq 0$), over all c in generation n , that have not appeared before. The resulting array is in OEIS as A226247 (denominators) and A226248 (numerators) [4, 5]. It is not surprising that *every* rational appears, since f and g generate the modular group $\mathrm{PSL}_2(\mathbb{Z}) \cong C_2 * C_3$ acting on the projective rational line [12, 13]. We prove it again directly in Theorem 3.7.

Writing $a(n)$ for the number of rationals appearing at the n -th step (with $a(0) = 1$ for the root 0), the observed values

$$1, 1, 2, 2, 3, 5, 7, 10, 15, 22, 32, 47, 69, 101, \dots$$

suggest the recurrence $a(n) = a(n - 1) + a(n - 3)$. Kimberling’s entry says, without a proof, that these counts are OEIS A097333 [7]. Kagey asked about this on Mathematics Stack Exchange in April 2025 [2], and it is Problem 137 of his Open Problem Collection [1], together with two related questions.

Q1. Is $a(n) = a(n-1) + a(n-3)$ for all $n \geq 4$?

Q2. Color a vertex **blue** if the last map applied to reach it is g , and **red** if it is f . Is a vertex blue if and only if its value is negative?

Q3. Is there a way to characterize the rank- n rational numbers?

The accepted answer to [2], by the user mathmasterzach [3], got most of the way on Q1. Using $g^2 = \text{id}$ and $(g \circ f)^3 = \text{id}$, it noted that shortest words in f and g have to avoid certain patterns. It then counted the words that avoid them with a 14-state transfer matrix and got the generating function $(1 + t^2 - t^3)/(1 - t - t^3)$. But that counts *words*, not *rationals*. The answer does not show that every word avoiding the patterns is a shortest word, that different such words end at different rationals, or that every rational is reached. Those are the hard steps, and we prove them here. Our main tool is a formula for the rank of a single rational, which also answers Q3.

We write $d(x)$ for the rank of x , the least number of steps from 0 to x (Definition 2.1). Rationals are always in lowest terms p/q with $q \geq 1$.

Theorem A (= Theorem 5.1). Let $a(n)$ be the number of rationals of rank n . Then

$$\sum_{n \geq 0} a(n) t^n = \frac{1 + t^2 - t^3}{1 - t - t^3}.$$

Hence $a(n) = a(n-1) + a(n-3)$ for all $n \geq 4$. It is false at $n = 3$, since $a(3) - a(2) - a(0) = -1$, so we cannot lower the 4. Also $a(n) = A097333(n-1)$ for $n \geq 1$, the number of *positive* rationals of rank n is Narayana's cows sequence A000930($n-1$), and

$$a(n) = C \psi^n + O(\psi^{-n/2}), \quad C = \frac{\psi^2 + \psi}{\psi^2 + 3} = 0.7019310679 \dots,$$

where $\psi = 1.4655712318 \dots$ (the “supergolden ratio”) is the real root of $\psi^3 = \psi^2 + 1$.

Theorem B (= Theorems 3.7, 4.3 and 4.9). Define the parent map $P(x) = x - 1$ for $x > 0$ and $P(x) = -1/x$ for $x < 0$. Then for every rational x the iteration of P reaches 0, and $d(x)$ is the number of steps it takes. More precisely,

(i) $d(0) = 0$, and $d(x) = d(-1/x) + 1$ for $x < 0$,

(ii) if $x > 0$ has negative continued fraction expansion $x = [c_1, \dots, c_k]^-$ with $c_1 \geq 1$ and $c_2, \dots, c_k \geq 2$ (Lemma 4.2), then

$$d(x) = c_1 + c_2 + \dots + c_k + k - 1,$$

(iii) if $x > 0$ is written as the unique ordinary continued fraction $x = [b_0; b_1, \dots, b_m]$ with m odd (Lemma 4.7), then

$$d(x) = \sum_{i \text{ even}} b_i + 3 \sum_{i \text{ odd}} b_i - 2.$$

In particular the rank- n rationals are enumerated by integer compositions (Corollary 4.4), answering Q3.

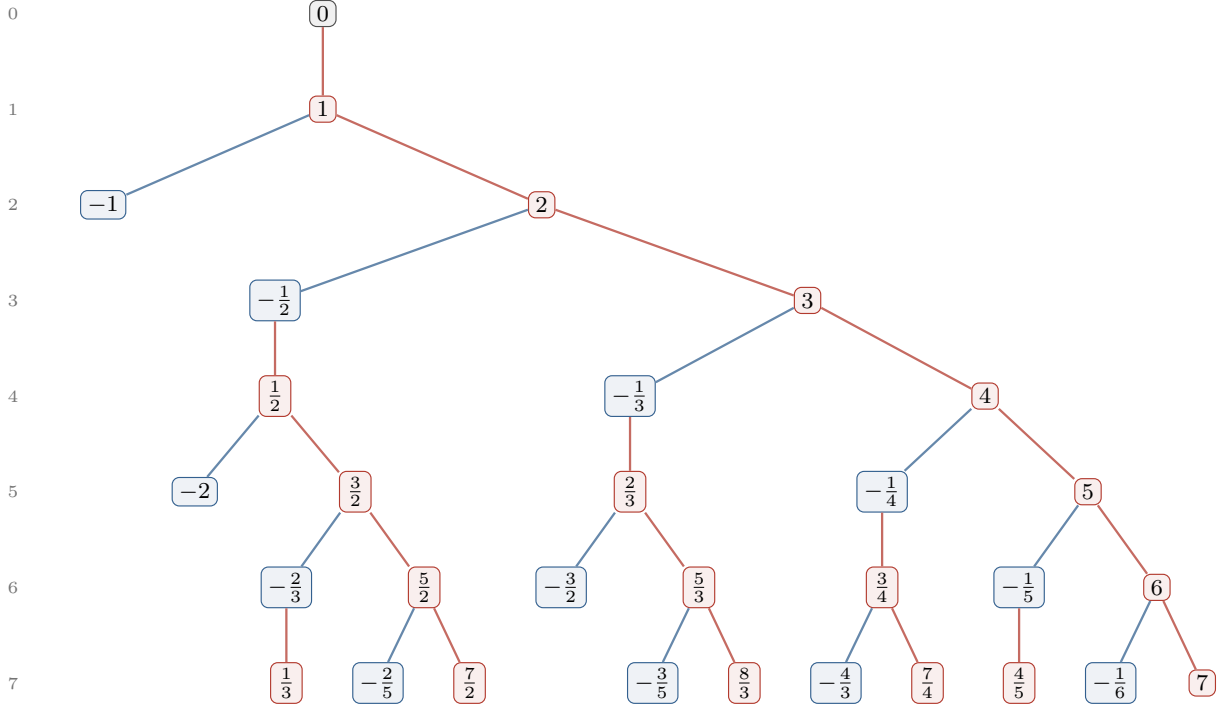


Figure 1: Ranks 0–7 of the tree. Red edges apply $f(x) = x + 1$ and blue edges apply $g(x) = -1/x$. Every positive vertex is entered by an f -edge and every negative vertex by a g -edge (Theorem C), and vertices ≤ -1 are leaves. The numbers on the left are ranks. The rows are Kimberling's generations [4], with his generation j being rank $j - 1$.

Theorem C (= Corollary 3.9). The only neighbour of $x \neq 0$ that is closer to the root is $P(x)$. So Kimberling's construction really is a tree, and no value shows up twice, not even twice in the same generation. The edge into x is an f -edge when $x > 0$ and a g -edge when $x < 0$, so *a vertex is blue if and only if its value is negative*, which answers Q2. A vertex $x > 0$ has two children, $x + 1$ and $-1/x$. A vertex $-1 < x < 0$ has one child, $x + 1$. A vertex $x \leq -1$ is a leaf, and for $x < -1$ the value $x + 1$ is not new but appeared exactly *three* ranks earlier.

The value $x + 1$ coming back three ranks early is what gives the $a(n - 3)$ term in the recurrence, and we will see that it comes from the relation $(gf)^3 = \text{id}$ in $\text{PSL}_2(\mathbb{Z})$.

Plan of the paper. Section 2 checks that Kimberling's generations, Kagey's figure and the sets X_n of [2] all give the same ranks. The main work is in Section 3, where we define the parent map and prove the distance formula (Theorem 3.7). The tree structure and Q2 follow right away. Section 4 puts the formula in terms of continued fractions to answer Q3, and Section 5 counts the ranks to prove Theorem A. In Section 6 we go back to the words counted in [3] and show they are exactly the shortest words. The last two sections have the computer checks and a few remarks. We do not rely on any of the references for the proofs.

2 The graph, ranks and generations

Definition 2.1. Let G be the directed graph on $\mathbb{Q}$ with an f -edge $x \rightarrow x + 1$ for every x , and a g -edge $x \rightarrow -1/x$ for every $x \neq 0$. There is no g -edge out of 0, since $g(0)$ is not defined. For

$x \in \mathbb{Q}$ let

$$d(x) = \text{length of a shortest directed path from } 0 \text{ to } x \text{ in } G \in \mathbb{Z}_{\geq 0} \cup \{\infty\},$$

and call $d(x)$ the *rank* of x . In other words, $d(x)$ is the smallest number of times we have to apply f and g to get from 0 to x , never applying g to 0. As in [2], we write

$$X_n = \{x \in \mathbb{Q} \mid d(x) = n\}, \quad a(n) = |X_n|.$$

Theorem 3.7 will show that $d(x) < \infty$ for all x .

Note that $d(0) = 0$ and $X_0 = \{0\}$, since the only path of length 0 is the empty one.

Kimberling [4] builds the rows as follows. Row 1 is (0). If row n is $(c_1, \dots, c_z)$, write down $(c_1 + 1, -1/c_1, c_2 + 1, -1/c_2, \dots)$, leaving out $-1/c_i$ when $c_i = 0$, and delete anything that is already in rows $1, \dots, n$. What is left is row $n + 1$. This is breadth-first search from 0, so his rows should be our sets X_n , and the next proposition checks this. Kagey's ranks [1] are the same as well.

Proposition 2.2 (generations are metric spheres). *For every $n \geq 0$, the set of values in Kimberling's row $n + 1$ equals X_n .*

Proof. We use induction on n . For $n = 0$ this is the definition. Suppose rows $1, \dots, n$ contain the values in $X_0, \dots, X_{n-1}$. Take y in row $n + 1$. Then y comes from some x in row n , that is $x \in X_{n-1}$, so $d(y) \leq n$. Since y was not deleted, it is not in $X_0, \dots, X_{n-1}$, so $d(y) \geq n$. Hence $y \in X_n$. Conversely let $y \in X_n$ with $n \geq 1$ and let x be the predecessor of y on a shortest path from 0. A prefix of a shortest path is a shortest path, so $d(x) = n - 1$ and x is in row n . So y is written down when we form row $n + 1$, and it is not deleted because $d(y) = n$. $\square$

Remark 2.3. There is one thing to be careful about. A value y could in principle be written down *twice* when forming row $n + 1$, once as $c + 1$ and once as $-1/c'$ for two different entries c, c' of row n , and the construction does not say what to do with the second copy. Corollary 3.9(a) shows that this never happens, because each $y \neq 0$ has only one in-neighbour at rank $d(y) - 1$. So the rows have no repeats and $|\text{row } n + 1| = a(n)$. Our computer check in Section 7 also matches the published A226247/A226248 data, including the order within each row.

Words. We will often write paths as words in f and g , read left to right in the order the maps are applied. For example fg sends 0 to $f(0) = 1$ and then to $g(1) = -1$. (As a composition, the word $\ell_1 \ell_2 \dots \ell_j$ is $\ell_j \circ \dots \circ \ell_1$.) We call a word *admissible at 0* if it never applies g to 0. The admissible words are the paths in G that start at 0, so $d(x)$ is the length of the shortest admissible word ending at x .

3 The parent map and the exact distance formula

We want a shortest path from 0 to each x that we can write down. Looking at Figure 1, it seems that a positive vertex is always entered by the f -edge from $x - 1$ and a negative vertex by the g -edge from $-1/x$. If that is right, then undoing the last step gives a “parent” for each vertex, and following parents should lead back to 0. This section proves it.

Definition 3.1. The *parent map* $P : \mathbb{Q} \setminus \{0\} \rightarrow \mathbb{Q}$ is

$$P(x) = \begin{cases} x - 1, & x > 0, \\ -1/x, & x < 0. \end{cases}$$

Note that P undoes an edge of G . For every $x \neq 0$ there is an edge $P(x) \rightarrow x$, which is the f -edge $(x - 1) \rightarrow x$ when $x > 0$ and the g -edge $(-1/x) \rightarrow x$ when $x < 0$. The second one is allowed because $-1/x \neq 0$.

Lemma 3.2 (termination). *For every $x \in \mathbb{Q}$ the iteration $x, P(x), P^2(x), \dots$ reaches 0 after finitely many steps.*

Proof. First let $x > 0$. Subtracting 1 a total of $\lceil x \rceil$ times gives $x - \lceil x \rceil$, which is 0 if $x \in \mathbb{Z}$ and is in $(-1, 0)$ otherwise. Next, if $z \in (-1, 0)$, write $z = -r/q$ in lowest terms with $0 < r < q$. Then $P(z) = -1/z = q/r$ is bigger than 1 and has the smaller denominator r .

Now we induct on the denominator q of x , for $x \in \mathbb{Q}_{>0} \cup (-1, 0)$. If $q = 1$, there is nothing in $(-1, 0)$ with denominator 1, and a positive integer x reaches 0 after x steps. If $q \geq 2$ and $x > 0$, then x is not an integer, so its orbit reaches a point of $(-1, 0)$ with the same denominator q . The next point is positive with denominator less than q , so we are done by induction. If $q \geq 2$ and $x \in (-1, 0)$, then $P(x)$ is positive with denominator less than q , and again we use induction.

Finally, if $x \leq -1$ then $P(x) = -1/x > 0$, which is the positive case, and $x = 0$ is clear. $\square$

Definition 3.3. For $x \in \mathbb{Q}$ let $D(x)$ be the number of steps in which the P -orbit of x reaches 0. Thus $D(0) = 0$ and

$$D(x) = D(P(x)) + 1 \quad (x \neq 0), \tag{1}$$

and these two properties determine D , as the next lemma shows.

Lemma 3.4 (uniqueness principle). *If $F : \mathbb{Q} \rightarrow \mathbb{Z}_{\geq 0}$ satisfies $F(0) = 0$ and $F(x) = F(P(x)) + 1$ for all $x \neq 0$, then $F = D$.*

Proof. By Lemma 3.2 the orbit $x, P(x), \dots, P^{D(x)}(x) = 0$ is finite. Using the recursion for F along it gives $F(x) = D(x) + F(0) = D(x)$. $\square$

Next we see how D changes along each of the four kinds of edges in G , including the two that P does not undo. The first three are easy, and (D4) is where the work is.

Lemma 3.5 (increment identities). *For all $x \in \mathbb{Q}$ the following hold.*

(D1) *If $x > 0$, then $D(x) = D(x - 1) + 1$.*

(D2) *If $x < 0$, then $D(x) = D(-1/x) + 1$.*

(D3) *If $x > 0$, then $D(-1/x) = D(x) + 1$.*

(D4) *If $x < 0$, then $D(x - 1) = D(x) + 3$.*

Proof. (D1) and (D2) are just (1). For (D3), $-1/x < 0$, so (D2) applied to $-1/x$ gives $D(-1/x) = D(-\frac{1}{-1/x}) + 1 = D(x) + 1$.

For (D4), put $z = x - 1$. Since $x < 0$, we have $z < -1$. We follow the orbit of z for four steps, checking the sign each time to see which case of P applies. The orbit is

$$z \xrightarrow{P} \frac{1}{1-x} \xrightarrow{P} \frac{x}{1-x} \xrightarrow{P} 1 - \frac{1}{x} \xrightarrow{P} -\frac{1}{x} = P(x).$$

Indeed, $z < 0$, so $P(z) = -1/z = 1/(1-x)$, which is in $(0, 1)$ because $1-x > 1$. Then $P^2(z) = \frac{1}{1-x} - 1 = \frac{x}{1-x}$, which is negative, and in fact in $(-1, 0)$ since it equals $-u/(1+u)$ with $u = -x > 0$. Then $P^3(z) = -\frac{1-x}{x} = 1 - \frac{1}{x}$, which is bigger than 1 because $-1/x > 0$. Finally $P^4(z) = -\frac{1}{x}$, and this is $P(x)$ because $x < 0$. Therefore $D(z) = 4 + D(P(x)) = 4 + (D(x) - 1) = D(x) + 3$. $\square$

Remark 3.6. Read backwards, the computation in (D4) gives the path

$$-\frac{1}{x} \xrightarrow{f} 1 - \frac{1}{x} \xrightarrow{g} \frac{x}{1-x} \xrightarrow{f} \frac{1}{1-x} \xrightarrow{g} x - 1,$$

so $x - 1 = (g \circ f \circ g \circ f \circ g)(x)$. This is the relation $f^{-1} = g \circ f \circ g \circ f \circ g$ from [1], or equivalently $(g \circ f)^3 = \text{id}$, which together with $g^2 = \text{id}$ presents $\text{PSL}_2(\mathbb{Z}) \cong C_2 * C_3$ [12, 13]. The 3 in (D4) is the 5 letters of the relator minus the 2 that cancel in $g \cdot g = \text{id}$, and this 3 is what later gives the t^3 in Theorem A. We did not need any group theory for the proof, though, since it is a direct check.

Theorem 3.7 (exact distance formula, dynamic form). *Every rational is reachable from 0 in G , and $d(x) = D(x)$ for all $x \in \mathbb{Q}$.*

Proof. Upper bound $d \leq D$. The reversed P -orbit $0 = P^{D(x)}(x)$, $P^{D(x)-1}(x)$, $\dots$, $P(x)$, x is a directed path in G , because each consecutive pair $(P(y), y)$ is an edge, as noted after Definition 3.1. It has length $D(x)$, so x is reachable and $d(x) \leq D(x)$.

Lower bound $d \geq D$. We first check that

$$D(x) \leq D(y) + 1 \quad \text{for every edge } y \rightarrow x \text{ of } G. \quad (2)$$

There are five cases.

- For an f -edge into $x > 0$, $y = x - 1$ and (D1) gives $D(x) = D(y) + 1$.
- For an f -edge into $x = 0$, $y = -1$ and $D(0) = 0 \leq D(-1) + 1$.
- For an f -edge into $x < 0$, $y = x - 1$ and (D4) gives $D(x) = D(y) - 3 \leq D(y) + 1$.
- For a g -edge into $x < 0$, $y = -1/x$ and (D2) gives $D(x) = D(y) + 1$.
- For a g -edge into $x > 0$, $y = -1/x < 0$ and (D3) gives $D(y) = D(x) + 1$, so $D(x) = D(y) - 1$.

There is no g -edge into 0. Now we induct on $n = d(x)$. For $n = 0$ we have $x = 0$ and $D(0) = 0$. For $n \geq 1$, let $y \rightarrow x$ be the last edge of a shortest path. Then $d(y) = n - 1$, so $D(y) \leq d(y)$ by induction, and (2) gives $D(x) \leq D(y) + 1 \leq d(y) + 1 = d(x)$. $\square$

From now on we use d and D interchangeably. The next two corollaries describe what the tree looks like near each vertex, and together they prove Theorem C.

Corollary 3.8 (in-neighbour distances). *Let $x \neq 0$. The in-neighbours of x in G are $x - 1$ (by f) and $-1/x$ (by g), with ranks as in the table.*

	in-neighbour $x - 1$	in-neighbour $-1/x$
$x > 0$	$d(x) - 1$	$d(x) + 1$
$x < 0$	$d(x) + 3$	$d(x) - 1$

In particular $P(x)$ is the only in-neighbour of x that is closer to the root than x .

Proof. These are the in-neighbours because $f^{-1}(x) = x - 1$ always exists and $g(y) = x$ only has the solution $y = -1/x$. The table is (D1)–(D4) together with Theorem 3.7. Note that $x - 1 \neq -1/x$ for rational x , since $x^2 - x + 1$ has no rational root, so the two in-neighbours are different. $\square$

Corollary 3.9 (tree structure, answering Q2). *(a) Each $x \neq 0$ is produced at stage $d(x)$ by exactly one element of $X_{d(x)-1}$, its parent $P(x)$. So no value is produced twice, not even twice in one generation, and the construction gives a rooted tree without any choices.*

(b) The edge into x is an f -edge when $x > 0$ and a g -edge when $x < 0$. With the colors of Q2, a vertex is blue if and only if its value is negative. The root has no color.

(c) A vertex $x > 0$ has the two children $x + 1$ and $-1/x$. A vertex $x \in (-1, 0)$ has the one child $x + 1 \in (0, 1)$. Every vertex $x \leq -1$ is a leaf. Moreover, for $x < -1$ the value $x + 1$ is not new but three ranks old, with $d(x + 1) = d(x) - 3$, and for $x = -1$ the value $x + 1 = 0$ is the root.

Proof. (a) When we form generation $d(y) + 1$, the value y is written down once for each of its in-neighbours in $X_{d(y)-1}$. By Corollary 3.8 there is only one, $P(y)$.

(b) The unique parent edge is the f -edge from $x - 1$ when $x > 0$ and the g -edge from $-1/x$ when $x < 0$.

(c) Let $x > 0$. Its out-neighbours are $x + 1$ and $-1/x$. By (D1) applied to $x + 1$, and by (D3), both have rank $d(x) + 1$, so both are children. Now let $x \in (-1, 0)$. Then $x + 1 \in (0, 1)$ is positive, and (D1) applied to $x + 1$ gives $d(x + 1) = d(x) + 1$, so it is a child, while $-1/x$ is the parent. Next let $x < -1$. Then $-1/x$ is the parent. Also $x = (x + 1) - 1$ with $x + 1 < 0$, so (D4) applied to $x + 1$ gives $d(x) = d(x + 1) + 3$. So $x + 1$ is three ranks older than x and is not a child. Finally, the out-neighbours of $x = -1$ are 0, the root, and 1, its parent, so it is a leaf. $\square$

Example 3.10. Take $x = -7/2$. The P -orbit is

$$-7/2 \rightarrow 2/7 \rightarrow -5/7 \rightarrow 7/5 \rightarrow 2/5 \rightarrow -3/5 \rightarrow 5/3 \rightarrow 2/3 \rightarrow -1/3 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0,$$

so $d(-7/2) = 12$. Reversing the orbit gives the shortest word $fffgffgffgfg$, which means we apply $f, f, f, g, f, f, g, f, f, g, f, g$ to 0 in this order.

4 Continued-fraction forms of the distance formula

Starting from a positive rational, P subtracts 1 a total of $\lceil x \rceil$ times and then flips. This is the algorithm for the *negative* (Hirzebruch–Jung, or “minus”) continued fraction [9, 10], so the length of the orbit can be read off from the expansion. We prove what we need from scratch.

Definition 4.1. For integers $c_1, \dots, c_k$ define, when the right sides are defined and nonzero where required,

$$[c_1]^- = c_1, \quad [c_1, c_2, \dots, c_k]^- = c_1 - \frac{1}{[c_2, \dots, c_k]^-}.$$

Lemma 4.2 (existence and uniqueness of the expansion). *Every rational $x > 0$ has exactly one representation*

$$x = [c_1, c_2, \dots, c_k]^-, \quad k \geq 1, \quad c_1 \geq 1, \quad c_i \geq 2 \quad (2 \leq i \leq k),$$

and in it $c_1 = \lceil x \rceil$. Moreover every tail satisfies $[c_j, \dots, c_k]^- > 1$ for $j \geq 2$.

Proof. Existence. Given $x > 0$, write down $c = \lceil x \rceil$. If $x = c$ we stop, and otherwise we repeat with $x' = 1/(c - x)$. If $x = p/q$ in lowest terms is not an integer, then $cq - p$ is between 1 and $q - 1$, and $x' = q/(cq - p)$. This is again in lowest terms, since a common divisor of q and $cq - p$ divides p , and its denominator $cq - p$ is less than q . So the process stops. Also $x' > 1$, so every digit after the first is $\lceil x' \rceil \geq 2$, and $c_1 = \lceil x \rceil \geq 1$ because $x > 0$. Using $x = c - 1/x'$ at each step shows that $x = [c_1, \dots, c_k]^-$.

Tails are bigger than 1. We use downward induction. First $[c_k]^- = c_k \geq 2 > 1$. If $y = [c_{j+1}, \dots, c_k]^- > 1$, then $[c_j, \dots, c_k]^- = c_j - 1/y > c_j - 1 \geq 1$.

Uniqueness. Suppose $x = [c_1, \dots, c_k]^-$ is any representation with these digit conditions. If $k = 1$ then $x = c_1 \in \mathbb{Z}$. If $k \geq 2$ then $x = c_1 - 1/y$ with $y = [c_2, \dots, c_k]^- > 1$ by the previous paragraph, so $x \in (c_1 - 1, c_1)$. Then x is not an integer, c_1 has to be $\lceil x \rceil$, and $y = 1/(c_1 - x)$ is fixed. By induction on k , an integer only has the representation with $k = 1$, and for a non-integer both the first digit and the tail are determined. $\square$

Theorem 4.3 (distance formula, negative-continued-fraction form). *Let $x > 0$ have the expansion of Lemma 4.2. Then*

$$d(x) = c_1 + c_2 + \dots + c_k + k - 1.$$

Together with $d(0) = 0$ and $d(x) = d(-1/x) + 1$ for $x < 0$, which is (D2), this gives d on all of $\mathbb{Q}$.

Proof. By Theorem 3.7 we can compute D instead. We induct on k . If $k = 1$ then $x = c_1$ is a positive integer, and its orbit is c_1 subtractions. Now let $k \geq 2$. Since $c_1 = \lceil x \rceil$, the orbit first subtracts 1 a total of c_1 times and gets to

$$x - c_1 = -\frac{1}{[c_2, \dots, c_k]^-} \in (-1, 0),$$

which is in $(-1, 0)$ because the tail is bigger than 1. One flip then gives $[c_2, \dots, c_k]^-$. By induction the rest of the orbit takes $(\sum_{i \geq 2} c_i) + (k - 1) - 1$ steps, so the total is $c_1 + 1 + \sum_{i \geq 2} c_i + k - 2 = \sum_i c_i + k - 1$. $\square$

Corollary 4.4 (the ranks explicitly, answering Q3). *For $n \geq 1$,*

$$\begin{aligned} X_n \cap \mathbb{Q}_{>0} &= \left\{ [c_1, \dots, c_k]^- \mid k \geq 1, c_1 \geq 1, c_2, \dots, c_k \geq 2, \sum_i c_i = n + 1 - k \right\}, \\ X_n \cap \mathbb{Q}_{<0} &= \left\{ -1/y \mid y \in X_{n-1} \cap \mathbb{Q}_{>0} \right\}, \end{aligned}$$

and in both lines each element is listed once, since different digit strings give different rationals and $y \mapsto -1/y$ is injective. In particular rank n is finite, and its elements correspond to integer compositions.

Proof. The first line follows from Lemma 4.2, which says each positive rational has exactly one digit string, and Theorem 4.3, which says its rank is $\sum c_i + k - 1$. For the second line, if $x < 0$ then $-1/x > 0$ and (D2) gives $d(-1/x) = d(x) - 1$. Conversely, if $y > 0$ then $-1/y < 0$ and (D3) gives $d(-1/y) = d(y) + 1$. So $x \mapsto -1/x$ is a bijection from $X_n \cap \mathbb{Q}_{<0}$ to $X_{n-1} \cap \mathbb{Q}_{>0}$. $\square$

Example 4.5. Take $n = 5$. The digit strings of weight 5 are (5) , $(1, 3)$ and $(2, 2)$, with values 5 , $[1, 3]^- = \frac{2}{3}$ and $[2, 2]^- = \frac{3}{2}$. The positive rationals of rank 4 are 4 and $\frac{1}{2}$, from the strings (4) and $(1, 2)$, and they give the negatives $-\frac{1}{4}$ and -2 . So $X_5 = \{5, \frac{2}{3}, \frac{3}{2}, -\frac{1}{4}, -2\}$, as in Figure 1.

Example 4.6. For $x = \frac{2}{7}$ the algorithm gives $\frac{2}{7} = [1, 2, 2, 3]^-$. To check, $[3]^- = 3$, $[2, 3]^- = \frac{5}{2}$, $[2, 2, 3]^- = \frac{7}{5}$ and $1 - \frac{5}{7} = \frac{2}{7}$. So $d(2/7) = (1 + 2 + 2 + 3) + 4 - 1 = 11$ and $d(-7/2) = 12$, as in Example 3.10.

4.1 The ordinary continued fraction form

Recall the ordinary (regular) continued fraction notation

$$[b_0; b_1, \dots, b_m] = b_0 + \frac{1}{b_1 + \frac{1}{\ddots + \frac{1}{b_m}}}, \quad b_0 \geq 0, \ b_i \geq 1 \ (1 \leq i \leq m), \ b_i \in \mathbb{Z}.$$

We call m the *length*. It is well known that a positive rational has exactly two such expansions (see [11, Ch. X]), but we prove the version we need to keep the paper self-contained.

Lemma 4.7 (odd-length normalization). *(a) For a digit string as above with $m \geq 1$, the value $v = [b_1; b_2, \dots, b_m]$ of the tail satisfies $v > 1$, except that $v = 1$ exactly when $m = 1$ and $b_1 = 1$.*

(b) Every rational $x > 0$ has exactly one expansion $x = [b_0; b_1, \dots, b_m]$ of odd length m .

(c) In the odd-length expansion, $b_0 \geq 1$ if and only if $x > 1$.

Proof. (a) If $m = 1$ then $v = b_1 \geq 1$ with equality iff $b_1 = 1$. If $m \geq 2$ then $v = b_1 + 1/[b_2; \dots; b_m] > b_1 \geq 1$.

(b) *Uniqueness.* We show by strong induction on the denominator that x has at most one expansion of each parity of m . Let $x = [b_0; b_1, \dots, b_m]$. If $m = 0$ then $x = b_0 \in \mathbb{Z}$. If $m \geq 1$, write $x = b_0 + 1/v$ with $v = [b_1; \dots, b_m]$. By (a) there are two cases. If $v > 1$, then $x \in (b_0, b_0 + 1)$ is not an integer, so $b_0 = \lfloor x \rfloor$ and $v = 1/(x - b_0)$ are forced, and the tail is an expansion of v of length $m - 1$. Otherwise $(m, b_1) = (1, 1)$ and $x = b_0 + 1 \in \mathbb{Z}$. So an integer $x \geq 1$ only has the expansions $[x]$ of length 0 and $[x - 1; 1]$ of length 1, one of each parity. For a non-integer x , the expansions of x of length m match up with the expansions of $v = 1/(x - \lfloor x \rfloor) > 1$ of length $m - 1$. Since v has a smaller denominator than x , it has at most one expansion of each parity by induction, and so does x .

Existence. The Euclidean algorithm gives one expansion. If its length has the wrong parity, we can change it by one using $[\dots, b_m] = [\dots, b_m - 1, 1]$ when $b_m \geq 2$, $[\dots, b_{m-1}, 1] = [\dots, b_{m-1} + 1]$ when $b_m = 1$, or $[x] = [x - 1; 1]$ for integers.

(c) Since m is odd, $m \geq 1$, so $x = b_0 + 1/v$ with $v \geq 1$. If $b_0 \geq 1$ then $x > b_0 \geq 1$, and if $b_0 = 0$ then $x = 1/v \leq 1$ by (a). Hence $b_0 \geq 1$ exactly when $x > 1$. Note that $x = 1$ itself has the odd-length form $[0; 1]$. $\square$

Lemma 4.8 (complement identity). *Let $b \geq 1$, let t be an admissible tail, possibly empty, and put $u = [b; t] \geq 1$. Then*

$$[0; 1, b, t] = \frac{u}{1 + u}, \quad [0; b + 1, t] = \frac{1}{1 + u},$$

and in particular $[0; 1, b, t] + [0; b + 1, t] = 1$.

Proof. We have $[1; b, t] = 1 + 1/u = (1 + u)/u$, so $[0; 1, b, t] = u/(1 + u)$. Also $[b + 1; t] = 1 + [b; t] = 1 + u$, so $[0; b + 1, t] = 1/(1 + u)$. $\square$

Theorem 4.9 (distance formula, ordinary-continued-fraction form). *Let $x > 0$ and let $x = [b_0; b_1, \dots, b_m]$ be its unique odd-length expansion (Lemma 4.7). Then*

$$d(x) = \sum_{i \text{ even}} b_i + 3 \sum_{i \text{ odd}} b_i - 2.$$

Proof. Let $E(x)$ be the right-hand side for $x > 0$, and extend E to all of $\mathbb{Q}$ by $E(0) = 0$ and $E(x) = E(-1/x) + 1$ for $x < 0$. By the uniqueness principle (Lemma 3.4) it is enough to show $E(x) = E(P(x)) + 1$ for every $x \neq 0$. For $x < 0$ this is the definition, so there are three cases left, all with $x > 0$.

Case $x > 1$. By Lemma 4.7(c), $b_0 \geq 1$, and $x - 1 = [b_0 - 1; b_1, \dots, b_m]$ is again an odd-length expansion, where the first digit may now be 0. By uniqueness it is *the* odd-length expansion of $x - 1$. Only the digit b_0 , in an even position, changed, so $E(x) = E(x - 1) + 1$.

Case $x = 1$. Here $E(1) = E([0; 1]) = 0 + 3 \cdot 1 - 2 = 1 = E(0) + 1$.

Case $0 < x < 1$. Here $b_0 = 0$ and $P(x) = x - 1 \in (-1, 0)$, so $E(x - 1) = E(\frac{1}{1-x}) + 1$, and we have to show $E(x) = E(\frac{1}{1-x}) + 2$. Note that $m \geq 1$, and if $b_1 = 1$ then $m \geq 3$, since $[0; 1] = 1$ is not in this case. We use Lemma 4.8.

- Suppose $b_1 = 1$, so $x = [0; 1, b_2, t]$ with tail $t = b_3, \dots, b_m$. Then $1 - x = [0; b_2 + 1, t]$, and hence

$$\frac{1}{1-x} = [b_2 + 1; b_3, \dots, b_m],$$

which is an expansion of odd length $m - 2$. Compared with the digits of x , position 0 went up by 1, which has weight 1, and the digit $b_1 = 1$ in an odd position is gone, which had weight 3. All the other digits moved by two places, so they kept the parity of their position. Hence $E(\frac{1}{1-x}) = E(x) + 1 - 3 = E(x) - 2$.

- Suppose $b_1 \geq 2$, so $x = [0; (b_1 - 1) + 1, t]$. Then $1 - x = [0; 1, b_1 - 1, t]$, and hence

$$\frac{1}{1-x} = [1; b_1 - 1, b_2, \dots, b_m],$$

which has the same odd length m . Position 0 went up by 1 and position 1 went down by 1, which has weight 3, and nothing else moved. Hence again $E(\frac{1}{1-x}) = E(x) + 1 - 3 = E(x) - 2$. $\square$

Example 4.10. The rational $\frac{2}{7} = [0; 3, 2]$ has odd-length form $[0; 3, 1, 1]$, so $d(2/7) = (0 + 1) + 3(3 + 1) - 2 = 11$, as in Example 4.6. Similarly $\frac{3}{5} = [0; 1, 1, 2]$ already has odd length, and $d(3/5) = (0 + 1) + 3(1 + 2) - 2 = 8$.

5 Counting the ranks

Theorem 5.1 (main count). *Let $r(n) = |X_n \cap \mathbb{Q}_{>0}|$. Then $a(0) = 1$, $a(n) = r(n) + r(n - 1)$ for $n \geq 1$, and, as formal power series,*

$$R(t) := \sum_{n \geq 1} r(n) t^n = \frac{t}{1 - t - t^3}, \quad A(t) := \sum_{n \geq 0} a(n) t^n = \frac{1 + t^2 - t^3}{1 - t - t^3}.$$

Consequently

(a) $a(n) = a(n - 1) + a(n - 3)$ for every $n \geq 4$,

(b) the recurrence fails at $n = 3$, where $a(3) - a(2) - a(0) = -1$, and

(c) $r(n) = A000930(n - 1)$ (Narayana's cows) and $a(n) = A097333(n - 1)$ for $n \geq 1$.

Proof. The equation $a(n) = r(n) + r(n-1)$ for $n \geq 1$ comes from the bijection in Corollary 4.4 between negatives of rank n and positives of rank $n-1$, and 0 is only in X_0 .

By Corollary 4.4, $r(n)$ is the number of digit strings $(c_1, \dots, c_k)$ with $c_1 \geq 1$ and $c_i \geq 2$ for $i \geq 2$, and with weight $\sum_i c_i + k - 1 = n$. We give a string the weight $t^{c_1} \prod_{i \geq 2} t^{c_i+1}$ and sum, which gives

$$R(t) = \sum_{k \geq 1} \left(\sum_{c \geq 1} t^c \right) \left(\sum_{c \geq 2} t^{c+1} \right)^{k-1} = \frac{t}{1-t} \sum_{k \geq 1} \left(\frac{t^3}{1-t} \right)^{k-1} = \frac{t}{1-t} \cdot \frac{1}{1 - \frac{t^3}{1-t}} = \frac{t}{1-t-t^3},$$

where the geometric series is fine as a formal power series because $t^3/(1-t)$ has no constant term. Hence

$$A(t) = 1 + R(t) + tR(t) = 1 + \frac{(1+t)t}{1-t-t^3} = \frac{(1-t-t^3) + t + t^2}{1-t-t^3} = \frac{1+t^2-t^3}{1-t-t^3},$$

using $\sum_{n \geq 1} r(n-1)t^n = tR(t)$, since $r(0) = 0$. Multiplying out gives $(1-t-t^3)A(t) = 1+t^2-t^3$, and comparing coefficients of t^n gives

$$a(n) - a(n-1) - a(n-3) = [t^n](1+t^2-t^3) = \begin{cases} 0, & n \geq 4, \\ -1, & n = 3, \end{cases}$$

which proves (a) and (b). For example $a(3) = 2$ while $a(2) + a(0) = 3$. For (c), A000930 has generating function $1/(1-t-t^3)$ and A097333 has generating function $(1+t)/(1-t-t^3)$ [8, 7], so $R(t) = t \cdot \frac{1}{1-t-t^3}$ and $A(t) - 1 = t \cdot \frac{1+t}{1-t-t^3}$ give the stated shifts. $\square$

Proposition 5.2 (asymptotics). *Let $\psi = 1.4655712318\dots$ be the real root of $\psi^3 = \psi^2 + 1$. Then*

$$a(n) = C \psi^n + O(\psi^{-n/2}), \quad C = \frac{\psi^2 + \psi}{\psi^2 + 3} = 0.7019310679\dots$$

Proof. The polynomial $x^3 - x^2 - 1$ has one real root $\psi > 1$ and a pair of complex conjugate roots $\lambda, \bar{\lambda}$. By Vieta the product of the three roots is 1, so $\psi |\lambda|^2 = 1$ and $|\lambda| = \psi^{-1/2} < 1$. Substituting $t = 1/x$ shows that the roots of $1-t-t^3$ are the simple roots $1/\psi$, $1/\lambda$ and $1/\bar{\lambda}$. So the pole of $A(t) = N(t)/(1-t-t^3)$, with $N(t) = 1+t^2-t^3$, closest to 0 is $t_0 = 1/\psi$. Using partial fractions,

$$a(n) = \frac{N(t_0)}{t_0(1+3t_0^2)} \psi^n + O(|\lambda|^n),$$

since the constant at the simple pole t_0 is $N(t_0)/(-Q'(t_0)t_0)$ with $Q(t) = 1-t-t^3$ and $-Q'(t) = 1+3t^2$. The other two poles have modulus $\psi^{1/2}$, so they give terms that shrink like $\psi^{-n/2}$. From $t_0^3 + t_0 - 1 = 0$ we get $N(t_0) = 1+t_0^2-t_0^3 = t_0 + t_0^2$, and hence

$$C = \frac{t_0 + t_0^2}{t_0(1+3t_0^2)} = \frac{1+t_0}{1+3t_0^2} = \frac{\psi^2 + \psi}{\psi^2 + 3},$$

multiplying numerator and denominator by ψ^2 and using $t_0 = 1/\psi$. $\square$

Proposition 5.3 (the four-class transfer structure). *For $n \geq 1$ put*

$$\alpha_n = |X_n \cap (1, \infty)|, \quad \beta_n = |X_n \cap (0, 1]|, \quad \gamma_n = |X_n \cap (-1, 0)|, \quad \delta_n = |X_n \cap (-\infty, -1]|.$$

Then for all $n \geq 1$,

$$\alpha_{n+1} = \alpha_n + \beta_n, \quad \beta_{n+1} = \gamma_n, \quad \gamma_{n+1} = \alpha_n, \quad \delta_{n+1} = \beta_n.$$

The transfer matrix has characteristic polynomial $t(t^3 - t^2 - 1)$.

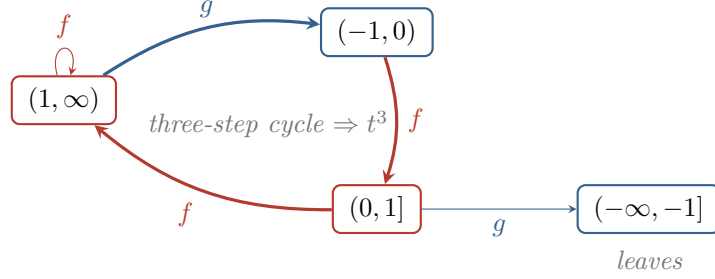


Figure 2: Where the children of each of the four classes go (Proposition 5.3). After a g -step, the only way back to the positives is the three-step cycle $(1, \infty) \rightarrow (-1, 0) \rightarrow (0, 1] \rightarrow (1, \infty)$. This cycle gives the t^3 in the recurrence, and it is the counting version of $(gf)^3 = \text{id}$ (Remark 3.6).

Proof. By Corollary 3.9, X_{n+1} is made of the children of X_n , and we just have to see where they land. For f -children, if $x > 1$ then $x + 1 > 2$, if $x \in (0, 1]$ then $x + 1 \in (1, 2]$, and if $x \in (-1, 0)$ then $x + 1 \in (0, 1]$. Vertices ≤ -1 have no children. So the elements of X_{n+1} in $(1, \infty)$ are the f -children of the positive vertices, and there are $\alpha_n + \beta_n$ of them, while the ones in $(0, 1]$ are the f -children of the vertices in $(-1, 0)$. Only positive vertices have g -children. If $x > 1$ then $-1/x \in (-1, 0)$, and if $x \in (0, 1]$ then $-1/x \in (-\infty, -1]$. This proves the four rules. The matrix of the linear map $(\alpha, \beta, \gamma, \delta) \mapsto (\alpha + \beta, \gamma, \alpha, \beta)$ is

$$M = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad \det(tI - M) = t(t^3 - t^2 - 1),$$

by expanding along the last column, which is zero because the δ -class has no children. □

6 Geodesic words, and completion of the transfer-matrix argument

If we reverse the P -orbit of a rational we get a shortest word for it. In this section we find the set of all these words and compare it with the words counted in [3].

Proposition 6.1 (canonical normal forms). *For $x \in \mathbb{Q}$ let $w(x)$ be the word we read along the reversed P -orbit of x , as in Example 3.10, with $w(0)$ empty. Then*

- (a) $w(x)$ is admissible at 0, ends at x , and has length $|w(x)| = d(x)$,
- (b) $x \mapsto w(x)$ is a bijection from $\mathbb{Q}$ onto the language

$$\mathcal{L} = \{\varepsilon\} \cup \left\{ f^{m_1} g f^{m_2} g \cdots f^{m_j} g^\epsilon \mid j \geq 1, \epsilon \in \{0, 1\}, m_1, \dots, m_{j-1} \geq 2, m_j \geq 1 \right\},$$

- (c) under this bijection $\epsilon = 1$ exactly when $x < 0$, and for $x > 0$ the run lengths are the digits of the negative continued fraction in reverse order, $(m_1, \dots, m_j) = (c_k, \dots, c_1)$.

Proof. Part (a) is the upper bound in the proof of Theorem 3.7, together with $d = D$. For (b) and (c), applying $w(x)$ to 0 gives back x , so the map is injective. Let $x > 0$ have digits $(c_1, \dots, c_k)$. By the proof of Theorem 4.3, its orbit is c_1 subtractions, a flip, c_2 subtractions, and so on, ending with c_k subtractions. Reversing this, the word is $f^{c_k} g f^{c_{k-1}} g \cdots g f^{c_1}$, so the runs are $(m_1, \dots, m_j) = (c_k, \dots, c_1)$ with $j = k$ and $\epsilon = 0$. The conditions $c_1 \geq 1$ and $c_i \geq 2$ for $i \geq 2$ become $m_j \geq 1$ and $m_i \geq 2$ for $i \leq j - 1$. For $x < 0$ we have $w(x) = w(-1/x) g$, which has the same runs and $\epsilon = 1$. Conversely, by Lemma 4.2 and Corollary 4.4, every choice of $(j, \epsilon, m_1, \dots, m_j)$ allowed in $\mathcal{L}$ comes from exactly one digit string and sign. $\square$

Proposition 6.2 (pattern-avoidance description). *$\mathcal{L}$ equals the set of words over $\{f, g\}$ containing no factor gg , containing no factor $fgfgf$, not beginning with g , and not beginning with fgf .*

Proof. A word with no gg that does not start with g can be written in exactly one way as

$$f^{m_1} g f^{m_2} g \cdots f^{m_j} g^\epsilon, \quad m_i \geq 1, \quad \epsilon \in \{0, 1\},$$

since the g 's in the middle are single and the word may end in g . In such a word, $fgfgf$ appears exactly when some run of length 1 has a g on each side and an f beyond each of those g 's. The run m_i has a g on each side when $2 \leq i \leq j - 1$, or when $i = j$ and $\epsilon = 1$, but in the second case nothing comes after the last g . So having no $fgfgf$ is the same as $m_i \geq 2$ for $2 \leq i \leq j - 1$. In the same way, the word starts with fgf exactly when $m_1 = 1$ and $j \geq 2$. Putting these together, the four conditions say $m_i \geq 2$ for $1 \leq i \leq j - 1$ and $m_j \geq 1$, which is the definition of $\mathcal{L}$. $\square$

Remark 6.3 (the Math.SE argument). The accepted answer [3] argued as follows. Since $g^2 = \text{id}$ and $fgfgf = g$, which follow from the relations in $\text{PSL}_2(\mathbb{Z})$, a *shortest* word cannot contain gg or $fgfgf$. A word also cannot start with g , which is not defined at 0, or with fgf , which goes back to 0. The answer then counted the words with these restrictions, which is the language $\mathcal{L}$ of Proposition 6.2, using a 14-state automaton, and got the generating function $(1 + t^2 - t^3)/(1 - t - t^3)$. This is correct and agrees with Theorem 5.1, but it does not show that the number of words of length n in $\mathcal{L}$ is $a(n)$. That is what Propositions 6.1 and 6.2 add. They show that *every* word in $\mathcal{L}$ is a shortest word, and not just that shortest words avoid the patterns. They also show that different words in $\mathcal{L}$ end at different rationals, and that every rational is reached. With this, the transfer matrix count is a second proof of Theorem 5.1(a). We also checked by computer that the tree words, $\mathcal{L}$ and the pattern-avoiding words are the same up to length 14.

7 Computational verification

We also ran a breadth-first search over $\mathbb{Q}$ in exact integer arithmetic, keeping fractions in lowest terms (the maps preserve this). This rebuilds Kimberling's construction up to rank 41, which is 14,144,885 vertices. We checked the following, and none of them failed.

1. The rank sizes $a(0), \dots, a(41)$ in Table 1 satisfy $a(n) = a(n - 1) + a(n - 3)$ for $4 \leq n \leq 41$, fail it at $n = 3$, and agree with A097333 shifted by one.
2. At every vertex, the edge in is an f -edge exactly when the value is positive (Q2).
3. At every vertex x , if $x > 0$ then $d(x - 1) = d(x) - 1$, if $x < 0$ then $d(-1/x) = d(x) - 1$, and if $x < -1$ then $d(x + 1) = d(x) - 3$. The numbers of children are 2, 1 and 0 for $x > 0$, $x \in (-1, 0)$ and $x \leq -1$, and no value is produced twice in a generation.

4. The rank formulas of Theorems 4.3 and 4.9 hold at all 453,457 vertices of rank at most 32, with the continued fractions computed separately from the search.
5. The rules of Proposition 5.3 hold at every rank.
6. The rows put together agree term by term with the published data of A226247 and A226248, including the order within each row (84 terms of each).

n	$a(n)$	n	$a(n)$	n	$a(n)$
0	1	14	148	28	31,224
1	1	15	217	29	45,761
2	2	16	318	30	67,066
3	2	17	466	31	98,290
4	3	18	683	32	144,051
5	5	19	1,001	33	211,117
6	7	20	1,467	34	309,407
7	10	21	2,150	35	453,458
8	15	22	3,151	36	664,575
9	22	23	4,618	37	973,982
10	32	24	6,768	38	1,427,440
11	47	25	9,919	39	2,092,015
12	69	26	14,537	40	3,065,997
13	101	27	21,305	41	4,493,437

Table 1: Verified rank sizes $a(n) = |X_n|$ for $0 \leq n \leq 41$.

8 Further remarks

Remark 8.1 (negative integers). The formulas are easy to use on special values. For $n \geq 1$, induction shows that $[2, 2, \dots, 2]^-$ with j twos is $(j+1)/j$. So $\frac{1}{n} = [1, 2^{(n-1)}]^-$, one 1 followed by $n-1$ twos, and $d(1/n) = (1 + 2(n-1)) + n - 1 = 3n - 2$ and

$$d(-n) = 3n - 1.$$

Thus $-1, -2, -3, \dots$ appear at ranks $2, 5, 8, \dots$, three apart, which again comes from the three-step cycle.

Remark 8.2 (sharpness at $n = 3$). The one failure, $a(3) = a(2) + a(0) - 1$, comes from the root. The word fgf would be allowed by the automaton of Section 6 if it did not go back to 0. Put another way, g is not defined at 0, so the tree has no vertex “ ∞ ”, and fgf is not allowed at the start. This removes one word of length 3, which is the $-t^3$ in the numerator of $A(t)$.

Remark 8.3 (positions of the nonpositive entries). If we list Kimberling’s rows one after another, the nonpositive values are at positions $n + A005374(n)$, where A005374 is Hofstadter’s H -sequence. This was noted in A226249 by Gómez Calderón (2025) [6]. We do not prove it here. It fits with Proposition 5.3, since the negatives of rank n match the positives of rank $n - 1$, and both counts grow by the same rule $t^3 = t^2 + 1$ that is behind A005374.

Remark 8.4 (variants). Kimberling’s entry also has a second ordering S' of the rationals, built from the same two maps with different conventions (A226130), and the matching count is conjectured in A226136 (see also the note in [2]). The parent-map method should work there too. We would first guess P from the small ranks and then prove the versions of (D1)–(D4).

Data and code availability

The verification script (`verify.py`, plain Python, ≈ 35 s) and the full solution notes are in the Problem 137 folder of <https://github.com/apemm/Kagey-Problems>.

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